

Table 1: Postfit nuisance pulls and saturated p -value for the 4D fit ($p_T^{\ell\ell} \times y^{\ell\ell} \times \cos \theta_{\ell\ell}^* \times \phi_{\ell\ell}^*$). Each NP cell is the physical postfit shift (rabbit pull $\times$ Nominal-config prior σ in physical units), so values are directly in the natural units of each NP (GeV²-like for λ 's, ν for the lattice γ parameters). Per-column $1\sigma_{\text{prior}}$ in the header is base kfactor $\times$ template half-range. `--scaleParams` factors in Scale5p0 configs are additionally absorbed into the cell so that a 1- σ pull in any config maps to the same physical shift. Δpull on α_S is relative to Nominal and scaled $\times 2$ to convert from the rabbit template step (0.001) to the analysis's quoted $1\sigma_{\alpha_S} = 0.002$ (matches `pullsAndImpacts.sh --scaleImpacts 2.0`); $\sigma(\alpha_S)$ is the absolute postfit total uncertainty in the same 10^{-3} units. The *Asimov* column is the postfit $\sigma(\alpha_S)$ from a separate `rabbit_fit -t -1` run on the same `ZMassDilepton.hdf5` (expected uncertainty under nominal-MC truth). Sat. p entries showing '—' are configurations where the saturated covariance Cholesky factorisation failed (Hessian not positive-definite) so the test could not be computed; the rest of the fit is unaffected.

Configuration	λ_2 (prefit = 0.25, $1\sigma_{\text{prior}} = 0.25$)	$\Delta\lambda_2$ (prefit = 0.125, $1\sigma_{\text{prior}} = 0.02$)	λ_4 (prefit = 0.06, $1\sigma_{\text{prior}} = 0.055$)	γ_{λ_2} (prefit = 0.087, $1\sigma_{\text{prior}} = 0.332$)	γ_{λ_2} (prefit = 0.0074, $1\sigma_{\text{prior}} = 0.066$)	γ_{λ_∞} (prefit = 1.6853, $1\sigma_{\text{prior}} = 5.069$)	f_{fake} (prefit = 0, $1\sigma_{\text{prior}} = 0.1$)	Sat. p	$\Delta\text{pull}(\alpha_S)$ [10^{-3}]	$\sigma(\alpha_S)$ [10^{-3}]	$\sigma(\alpha_S)$ Asimov [10^{-3}]
Nominal	+0.450 $\pm$ 0.235	−0.013 $\pm$ 0.007	−0.061 $\pm$ 0.051	+0.306 $\pm$ 0.111	−0.011 $\pm$ 0.025	−1.022 $\pm$ 5.057	—	2.54%	ref	1.103	1.089
Nominal + fakelumi	+0.304 $\pm$ 0.221	−0.014 $\pm$ 0.007	−0.045 $\pm$ 0.051	+0.272 $\pm$ 0.103	−0.022 $\pm$ 0.025	−0.850 $\pm$ 5.058	−0.085 $\pm$ 0.036	7.18%	+2.33	1.471	1.521
$\Lambda_4 \times 5$	+0.346 $\pm$ 0.215	−0.013 $\pm$ 0.007	−0.332 $\pm$ 0.120	+0.227 $\pm$ 0.105	−0.086 $\pm$ 0.039	−0.603 $\pm$ 5.060	—	7.64%	−0.44	1.094	1.107
$\Lambda_4 \times 5$ + fakelumi	+0.259 $\pm$ 0.217	−0.013 $\pm$ 0.007	−0.266 $\pm$ 0.126	+0.217 $\pm$ 0.104	−0.079 $\pm$ 0.039	−0.537 $\pm$ 5.060	−0.064 $\pm$ 0.037	12.41%	+1.43	1.543	1.610
$\Lambda_8 \times 5$	+1.102 $\pm$ 0.352	−0.015 $\pm$ 0.008	−0.248 $\pm$ 0.124	+0.589 $\pm$ 0.171	−0.105 $\pm$ 0.040	−0.717 $\pm$ 5.060	—	26.00%	−1.91	1.226	1.249
$\Lambda_8 \times 5$ + fakelumi	+0.961 $\pm$ 0.388	−0.015 $\pm$ 0.008	−0.222 $\pm$ 0.128	+0.538 $\pm$ 0.181	−0.098 $\pm$ 0.040	−0.667 $\pm$ 5.060	−0.035 $\pm$ 0.040	24.43%	−0.70	1.846	1.918
all NP asym.	+0.448 $\pm$ 0.179	−0.013 $\pm$ 0.007	−0.009 $\pm$ 0.028	+0.304 $\pm$ 0.084	−0.050 $\pm$ 0.036	−0.660 $\pm$ 4.100	—	2.75%	−0.15	1.080	1.089
all NP asym. + fakelumi	+0.315 $\pm$ 0.190	−0.013 $\pm$ 0.007	−0.008 $\pm$ 0.029	+0.274 $\pm$ 0.085	−0.052 $\pm$ 0.036	−0.605 $\pm$ 4.154	−0.079 $\pm$ 0.036	6.76%	+2.04	1.503	1.521
all NP $\times 5$	+1.747 $\pm$ 0.429	−0.014 $\pm$ 0.008	−0.320 $\pm$ 0.150	+0.892 $\pm$ 0.207	−0.173 $\pm$ 0.053	−7.09 $\pm$ 24.88	—	63.25%	−3.12	1.302	1.323
all NP $\times 5$ + fakelumi	+1.716 $\pm$ 0.481	−0.014 $\pm$ 0.008	−0.316 $\pm$ 0.154	+0.880 $\pm$ 0.223	−0.172 $\pm$ 0.054	−6.98 $\pm$ 24.89	−0.006 $\pm$ 0.041	—	−2.91	1.989	2.087
all NP $\times 5$, γ_{Λ_∞} default	+1.744 $\pm$ 0.429	−0.014 $\pm$ 0.008	−0.331 $\pm$ 0.145	+0.891 $\pm$ 0.207	−0.178 $\pm$ 0.050	−0.294 $\pm$ 5.065	—	63.05%	−3.11	1.302	1.323
all NP $\times 5$, γ_{Λ_∞} default + fakelumi	+1.712 $\pm$ 0.481	−0.014 $\pm$ 0.008	−0.326 $\pm$ 0.149	+0.878 $\pm$ 0.223	−0.176 $\pm$ 0.051	−0.289 $\pm$ 5.065	−0.006 $\pm$ 0.041	63.15%	−2.89	1.987	2.085
all NP $\times 2$	+1.099 $\pm$ 0.331	−0.014 $\pm$ 0.008	−0.155 $\pm$ 0.091	+0.602 $\pm$ 0.158	−0.080 $\pm$ 0.036	−3.00 $\pm$ 10.07	—	15.32%	−1.51	1.196	1.204
all NP $\times 2$ + fakelumi	+0.909 $\pm$ 0.357	−0.015 $\pm$ 0.008	−0.133 $\pm$ 0.092	+0.535 $\pm$ 0.165	−0.077 $\pm$ 0.036	−2.60 $\pm$ 10.07	−0.053 $\pm$ 0.038	20.30%	+0.22	1.726	1.777
all NP unconstrained	+0.843 $\pm$ 0.075	−0.014 $\pm$ 0.008	−0.878 $\pm$ 0.526	+0.436 $\pm$ 0.048	−0.339 $\pm$ 0.231	+152.83 $\pm$ 240.10	—	60.17%	−1.83	1.068	1.465
all NP unconstrained + fakelumi	+0.839 $\pm$ 0.078	−0.014 $\pm$ 0.008	−0.765 $\pm$ 0.558	+0.455 $\pm$ 0.059	−0.306 $\pm$ 0.237	+127.66 $\pm$ 243.74	−0.034 $\pm$ 0.044	—	−0.79	1.937	2.331
all NP unconstr., γ_{Λ_∞} default	+0.844 $\pm$ 0.076	−0.014 $\pm$ 0.008	−0.560 $\pm$ 0.165	+0.446 $\pm$ 0.045	−0.196 $\pm$ 0.055	+0.068 $\pm$ 5.068	—	58.41%	−1.71	1.053	1.365
all NP unconstr., γ_{Λ_∞} default + fakelumi	+0.841 $\pm$ 0.077	−0.015 $\pm$ 0.008	−0.489 $\pm$ 0.185	+0.467 $\pm$ 0.052	−0.185 $\pm$ 0.056	+0.055 $\pm$ 5.068	−0.037 $\pm$ 0.041	—	−0.54	1.737	2.212

Table 2: Postfit nuisance pulls and saturated p -value for the 1D fit ($p_T^{\ell\ell}$). Each NP cell is the physical postfit shift (rabbit pull $\times$ Nominal-config prior σ in physical units), so values are directly in the natural units of each NP (GeV²-like for λ 's, ν for the lattice γ parameters). Per-column $1\sigma_{\text{prior}}$ in the header is base kfactor $\times$ template half-range. --scaleParams factors in Scale5p0 configs are additionally absorbed into the cell so that a 1- σ pull in any config maps to the same physical shift.

Configuration	λ_2 (prefit = 0.25, $1\sigma_{\text{prior}} = 0.25$)	$\Delta\lambda_2$ (prefit = 0.125, $1\sigma_{\text{prior}} = 0.02$)	λ_4 (prefit = 0.06, $1\sigma_{\text{prior}} = 0.055$)	γ_{λ_2} (prefit = 0.087, $1\sigma_{\text{prior}} = 0.332$)	γ_{λ_4} (prefit = 0.0074, $1\sigma_{\text{prior}} = 0.066$)	γ_{Λ_∞} (prefit = 1.6853, $1\sigma_{\text{prior}} = 5.069$)	f_{fake} (prefit = 0, $1\sigma_{\text{prior}} = 0.1$)	Sat. p
Nominal	+0.315 $\pm$ 0.216	+0.004 $\pm$ 0.020	-0.045 $\pm$ 0.051	+0.264 $\pm$ 0.104	-0.004 $\pm$ 0.026	-0.750 $\pm$ 5.060	—	12.18%
Nominal + fakelumi	+0.156 $\pm$ 0.225	+0.002 $\pm$ 0.020	-0.026 $\pm$ 0.052	+0.236 $\pm$ 0.105	-0.024 $\pm$ 0.027	-0.554 $\pm$ 5.060	-0.117 $\pm$ 0.047	25.44%
$\Lambda_4 \times 5$	+0.249 $\pm$ 0.218	+0.003 $\pm$ 0.020	-0.284 $\pm$ 0.129	+0.201 $\pm$ 0.109	-0.069 $\pm$ 0.041	-0.496 $\pm$ 5.061	—	22.48%
$\Lambda_4 \times 5$ + fakelumi	—	—	—	—	—	—	—	—
$\Lambda_5 \times 5$	+0.902 $\pm$ 0.437	+0.021 $\pm$ 0.099	-0.215 $\pm$ 0.135	+0.523 $\pm$ 0.203	-0.087 $\pm$ 0.042	-0.588 $\pm$ 5.061	—	35.43%
$\Lambda_5 \times 5$ + fakelumi	+0.631 $\pm$ 0.493	+0.016 $\pm$ 0.099	-0.169 $\pm$ 0.141	+0.426 $\pm$ 0.219	-0.078 $\pm$ 0.043	-0.509 $\pm$ 5.062	-0.066 $\pm$ 0.056	36.94%
all NP asym.	+0.306 $\pm$ 0.204	+0.004 $\pm$ 0.021	-0.009 $\pm$ 0.031	+0.256 $\pm$ 0.097	-0.031 $\pm$ 0.042	-0.549 $\pm$ 4.221	—	11.82%
all NP asym. + fakelumi	—	—	—	—	—	—	—	—
all NP $\times 5$	+1.761 $\pm$ 0.552	+0.024 $\pm$ 0.099	-0.281 $\pm$ 0.163	+0.932 $\pm$ 0.260	-0.167 $\pm$ 0.059	-6.30 $\pm$ 24.95	—	69.75%
all NP $\times 5$ + fakelumi	—	—	—	—	—	—	—	—
all NP $\times 5$, γ_{Λ_∞} default	—	—	—	—	—	—	—	—
all NP $\times 5$, γ_{Λ_∞} default + fakelumi	—	—	—	—	—	—	—	—
all NP $\times 2$	—	—	—	—	—	—	—	—
all NP $\times 2$ + fakelumi	—	—	—	—	—	—	—	—
all NP unconstrained	+4.986 $\pm$ 3.983	-1.454 $\pm$ 2.588	+1.031 $\pm$ 1.454	+1.672 $\pm$ 0.612	+0.471 $\pm$ 0.638	-797.57 $\pm$ 712.91	—	71.38%
all NP unconstrained + fakelumi	—	—	—	—	—	—	—	—
all NP unconstr., γ_{Λ_∞} default	—	—	—	—	—	—	—	—
all NP unconstr., γ_{Λ_∞} default + fakelumi	—	—	—	—	—	—	—	—